SERA — An Ethically-Governed Digital Twin of the Italian Provinces: Comparing Distributive-Justice Objectives in AI Policy Optimization

Ethics in Artificial Intelligence

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Abstract

This project studies the ethical governance of an AI system that advises on the allocation of public resources. SERA (Socio-Economic Resource Allocator) is a *digital twin* of the 110 Italian provinces: it ingests ∼88 historical socioeconomic indicators (2001–2025) from the Italian National Institute of Statistics (ISTAT), trains one machine learning model per indicator, and simulates the provinces forward year by year under user-chosen policy levers (tax rates, spending allocations, regulatory settings) subject to a national budget constraint. On top of the twin, an AI policy optimizer searches for multi-year lever settings — but *what* it maximizes is an explicit, user-facing ethical choice among four pluggable objectives drawn from theories of distributive justice: Utilitarianism (total GDP), Rawlsian justice (maximin), Egalitarianism (Sen welfare), and a multi-indicator wellbeing composite. The optimizer itself is also a choice, spanning a deliberate *explainability spectrum* from white-box rule lists to a black-box neural policy audited post hoc, so that the price of transparency is measured rather than assumed. A multi-objective Pareto-frontier search (NSGA-II) reframes the efficiency–equity trade-off as a continuum instead of a dropdown. The system is wrapped in responsible-AI scaffolding — ethics statement, model card, datasheet, sensitivity bands, and a human-in-the-loop candidate/adopt workflow — and positioned explicitly with respect to the EU AI Act. The report includes an experimental comparison in which the same neural policy model is trained once per ethical framework from the same 2025 starting state and rolled out to 2035, replicated across several seeds: the frameworks drive the same optimizer to measurably different futures — the distribution-blind utilitarian run grows the average most but leaves the worst-off province the lowest of the interventions, while maximin (and a smoothed Conditional-Value-at-Risk variant added to test its sparse signal) reliably raises the floor and reproduces the leveling-down objection by cutting the richest tail, and Sen welfare buys the highest floor among the growth-positive options. A methodological result falls out of the replication: a single seed had shown maximin's floor falling *below* the utilitarian one — a tidy story that dissolved under replication as search luck. The NSGA-II search then reveals that *uniform* national policy cannot trade efficiency against inequality at all, while a clustered search that can target regions makes the trade-off reappear: distributional outcomes are determined by per-province *targeting*. Read in the dimension that motivated the choice of Italy — the North–South divide — the frameworks separate cleanly: the utilitarian tide leaves the gap untouched, maximin closes it by leveling the North down, and only Sen welfare narrows it while both regions grow. The project's central claim is that an optimizer always embodies an ethical framework; SERA's contribution is to make that framework visible, swappable, and inspectable.

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1 Introduction

Artificial Intelligence systems are increasingly consulted in decisions that allocate resources among people: budget planning, service provisioning, urban management, welfare administration. Whenever such a system recommends *who gets what*, it is not merely performing a technical optimization — it is enacting a theory of distributive justice, whether its designers acknowledge it or not. An optimizer that maximizes total output is a utilitarian; one that protects the worst-off is Rawlsian; one that penalizes inequality is egalitarian. The objective function is a value judgment wearing the costume of mathematics, and the costume is effective: a number that emerges from a pipeline of models reads as neutral in a way that the same judgment, stated in words, never would.

This project confronts that observation directly. SERA (the name plays on the Italian *sera* and is the acronym for "Socio-Economic Resource Allocator") is a digital twin of the Italian provinces. Italy is an ideal testbed for distributive questions: the North–South divide is one of the most persistent regional inequalities in Europe, and any national policy implicitly chooses between aggregate growth and territorial cohesion. SERA simulates the socioeconomic trajectories of the 110 provinces under user-controlled policy levers, and lets an AI optimizer propose lever settings over a multi-year horizon. Crucially, the system separates three questions that are usually conflated:

1. **What does the world do?** — the twin: per-indicator ML models plus a causal graph that is hand-written but *panel-reviewed* (its edge directions checked against, and corrected by, the historical data; Section 4.4), with honestly documented uncertainty.
2. **How do we search?** — the policy model: seven optimizers spanning an explainability spectrum from printable IF/THEN rules to a neural network.
3. **What counts as better?** — the ethical objective: four pluggable formalizations of distributive justice, plus a Pareto frontier that searches all of them at once.

Because any policy model can be paired with any objective, the distributional consequences of each ethical framework become directly comparable on the same model of the same country: what Utilitarianism does to the map, what Rawlsian maximin is willing to sacrifice, and how Sen welfare negotiates between them become empirical questions rather than positions. This report carries out that comparison experimentally (Section 9): the same neural policy is trained once per framework from the same starting state, and the resulting futures are compared on efficiency, inequality, and the floor — with results that match the textbook in some places and instructively depart from it in others.

The challenge this prototype abstracts is practical and current. Key real-world analogues include:

- **Regional development funds**: allocating EU cohesion or national recovery funds (e.g., Italy's PNRR) across territories, where the tension between efficiency and territorial equity is the explicit subject of political debate.
- **Public-service capacity planning**: distributing healthcare or education investment across regions with different returns and different needs.
- **Algorithmic benefit administration**: any scoring system that prioritizes applicants or regions for limited funds embodies the same choice between total measured benefit and protection of the worst placed.

SERA is decision *support*, never decision *making*: the optimizer returns graded candidates with stated trade-offs, and a human must explicitly adopt one — including, as a first-class option, doing nothing. The report proceeds as follows. Section 2 places the project in the machine-ethics landscape. Sections 3–4 present the technologies and the system design. Section 5 analyzes the implemented ethical frameworks philosophically and algorithmically, including the frameworks considered but not implemented. Section 6 details the policy-model spectrum and its mathematics; Section 7 discusses explainability as an ethical design axis; Section 8 covers human oversight, uncertainty, documentation, and the EU AI Act. Section 9 presents the experimental comparison of the four frameworks and the Pareto frontier; Section 10 discusses the findings and their limits; Section 11 concludes. Reproducibility instructions are in Section 12, and the appendix lists the core algorithms.

2 Background: Machine Ethics and Value-Laden Optimization

Moor's influential taxonomy of machine ethics [18] distinguishes *ethical-impact agents* (any machine whose use has ethical consequences), *implicit ethical agents* (machines whose safe behavior is built into their design), and *explicit ethical agents* (machines that represent ethical categories and reason over them). Most deployed optimization systems are ethical-impact agents pretending to be neither: their objective function encodes a moral stance — usually an unexamined sum-maximization — without representing it as such. SERA's design goal is to move one deliberate step up Moor's ladder: the ethical framework is reified as a first-class software object (an `Objective` class with an identifier, a label, and an honest natural-language description of what it ignores), selected by a human before any optimization runs. The machine does not *reason* about ethics — the choice of framework remains human — but the framework is explicit, inspectable, and swappable, rather than implicit in a hard-coded loss.

Three adjacent literatures inform the design. First, the *fairness in machine learning* literature has established that fairness admits multiple, mutually incompatible formalizations and that the choice among them is normative, not technical; SERA transposes that lesson from classification (individual versus group fairness) to resource allocation (utilitarian versus maximin versus inequality-discounted welfare). Second, the *AI governance* literature converges on a small set of principles — transparency, justice, non-maleficence, responsibility, privacy [19] — which SERA operationalizes as concrete artifacts: explainability badges, an ethics statement, a model card [21], a datasheet [22], and an explicit EU AI Act mapping [27]. Third, the *beyond-GDP* movement in welfare economics [17] supplies the critique that motivates the fourth objective: even a perfectly fair distribution of the wrong quantity is the wrong target.

A note on what SERA is not. It is not an attempt to build an artificial moral agent, nor to learn ethics from data: the frameworks are classical theories of distributive justice, hand-formalized, and the system's contribution is comparative infrastructure — the ability to hold everything else fixed while swapping the moral criterion, which philosophy can rarely do and policy experiments can never do.

3 Employed Technologies

3.1 Python and the scientific stack

The data pipeline and the twin engine are implemented in Python 3.11. The indicator models use `scikit-learn` (ridge regression by default, random forests as an alternative); the simulator, the gradient-free policy optimizers, and the NSGA-II search are written in `NumPy`/`pandas` with no deep-learning framework — the "neural" policy is a small pure-NumPy multilayer perceptron, trained by evolution strategies rather than backpropagation, since the twin it must optimize through is a non-differentiable composite of sklearn models and hand-written rules. Trained models are persisted with `joblib`.

3.2 The ISTAT SDMX API

All historical data comes from the official SDMX API of ISTAT, the Italian National Institute of Statistics. The client (`sera.istat_client`) is rate-limited — ISTAT enforces a hard limit of 5 requests/minute per IP, and exceeding it can lead to multi-day IP bans, so the client keeps a 25-second minimum delay between requests, counting failed requests too — and caches every response keyed by dataflow, dimensional key, and year range, so that a given data snapshot is preserved even though ISTAT revises history. A known quirk of the API (the `endPeriod` parameter returns year+1) is compensated for centrally. Each downloaded series carries a `.mapping.json` sidecar recording exactly which ISTAT dataflow and dimensions produced it, keeping every number traceable to its official source — a provenance requirement that the datasheet (Section 8.3) elevates from engineering hygiene to data governance.

3.3 Electron + React control-room UI

The user interface is an Electron desktop application with a React front end: a clickable province map (GeoJSON choropleth), per-province policy allocators, a national budget meter, the optimizer panel, an ethics equity dashboard, and the Pareto-frontier view. All JavaScript libraries are vendored; the app makes no network requests at runtime. The Electron main process talks to Python through a bridge script (`ui/backend_bridge.py`), one process per command (`bootstrap`, `province-trends`, `simulate-next-year`, `optimize-policy`, `compare-objectives`, `pareto-front`), streaming structured progress messages over `stderr`.

3.4 Pytest

The mechanics of the system — indicator bounds, the budget constraint, objective arithmetic, Pareto dominance, determinism of the policy optimizers, province-code normalization — are covered by a `pytest` suite. The tests deliberately cover *mechanics, not predictive accuracy*: asserting that the twin forecasts Italy correctly would be a claim the project explicitly refuses to make (Section 8.2). The tests are run automatically on every commit by GitHub Actions and gitlab CI, and the system refuses to run if any test fails.

4 The System

4.1 Project structure

- `src/sera/` — the Python package.
  - `config.py` — paths, ISTAT constants, the indicator→category map.
  - `istat_client.py`, `downloader.py`, `downloaders/<category>/<indicator>.py` — the rate-limited, cached data pipeline; one module per indicator.
  - `twin/` — the digital twin:
    - `data_loader.py` — panel assembly, disaggregation, interpolation;
    - `model_trainer.py` — one regressor per indicator;
    - `causal_graph.py` — 20 policy levers, their documented lever→indicator links, inter-indicator dependencies, bounds and effect directions;
    - `simulator.py` — the year-step engine;
    - `budget.py` — the national budget constraint (pool, tax funding, reserve carry-over), a pure package module so the headless experiments need not import the UI layer;
    - `policy.py` — the rollout environment, the budget constraint seam, and the seven policy models;

    * `objectives.py` — the seven ethical objectives;
    * `explain.py` — post-hoc audits for black-box policies;
    * `pareto.py` — the NSGA-II efficiency–equity frontier (uniform or per-cluster);
    * `panel_estimation.py` — fixed-effects panel estimation that re-derives the twin's structure from the data, scores it against the hand-written graph, and produces the reviewed edge signs the simulator consumes (Section 4.4);
    * `cli.py` — training and batch simulation.

- `ui/` — the Electron app (`main.js`, `preload.js`, `renderer.js`, `backend_bridge.py`).
- `data/` — downloaded indicator CSVs (110 provinces, 2-letter sigle).
- `ETHICS.md`, `docs/MODEL_CARD.md`, `docs/DATASHEET.md` — the responsible-AI documentation (Section 8.3).
- `tests/` — the pytest suite.
- `report/` — this report, with the experiment scripts (`run_experiments.py`, `plot_results.py`) that generated Section 9.

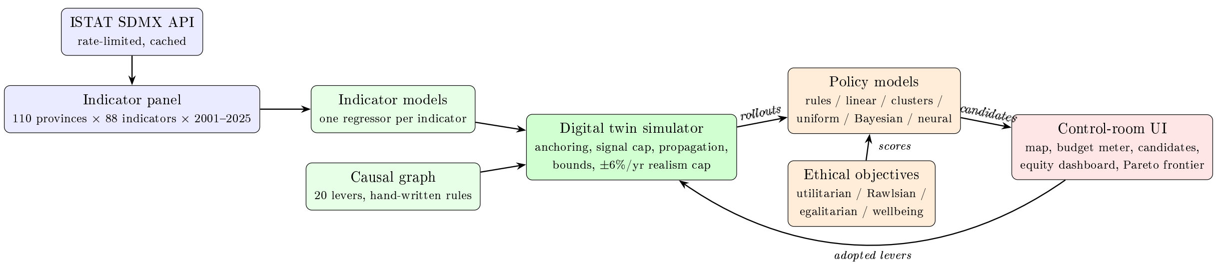

Figure 1: SERA architecture. The ethical objective is a separate, swappable component: the search procedure (policy model) and the moral criterion (objective) are independent choices, both exposed in the UI.

4.2 The data pipeline and its ethical caveats

The downloader fetches ∼88 indicators across economic, labor, education, healthcare, social-wellbeing, environment, energy, innovation, transportation, and public-finance categories. The loaded panel is the unit "province–year": 110 provinces identified by two-letter sigle, 2001–2025, with province codes normalized across Italy's post-reform administrative changes (`sera.twin.province_mapping`). The UI and the experiments in this report work on a 24-indicator subset spanning the economic, labor, education, healthcare, environmental, mobility, and social-wellbeing domains.

Following the practice of *Datasheets for Datasets* [22], the project documents the preprocessing that makes the panel usable and the distortions it introduces:

- **Disaggregation, and a provenance label for it.** Indicators published only at national or regional level are spread down to provinces on a population/share basis. Those provincial values are *estimates*, not measurements; fine-grained provincial variation for such indicators is partly an artifact. The loader now classifies each indicator's *provenance* (`DataLoader.classify_provenance`) by inspecting the raw file's area codes: of the 24 indicators in the experiment panel, only 11 are directly *measured* at province level, while 6 are disaggregated from regional and 7 from national data — 13 of 24 in total. Critically, `gdp_per_capita` itself is disaggregated regional, so the entire GDP-denominated equity analysis rests on disaggregated provincial values: a Rawlsian run chasing the "worst-off province" may be chasing a population-share artifact as much as a real place. The provenance map makes this quantifiable rather than a footnote, and is the prerequisite for the future work of down-weighting disaggregated dimensions in the objectives.

- **Interpolation.** Missing province-years are interpolated when standardizing to the 110-province panel. When the simulation needs a baseline year for which an indicator has no data, the most recent available year is used and logged (in the experiments of Section 9, most indicators resolve to 2022–2024; two slow-moving infrastructure series fall back to 2012).
- **Who is missing.** Official statistics measure the officially visible. The informal economy (significant and regionally uneven in Italy), undocumented residents, the homeless, and institutionalized populations are under- or un-represented. *Any optimization on this data optimizes for the measured population* — a primary ethical caveat of the whole project, stated in the datasheet and repeated in the ethics statement.

The last point deserves emphasis because it interacts with the choice of ethical objective. A Rawlsian objective directs resources to the worst-off *measured* province; if the worst-off people are systematically under-measured (as the informal economy and undocumented populations are), even the most equity-oriented framework inherits the measurement's blind spots. No objective function can correct for people the data cannot see.

4.3 The digital twin

The twin advances the provincial state one year at a time (Algorithm 1 in the appendix). One trained regressor per indicator predicts next-year values from lagged indicators plus the year's policy levers; but the design deliberately does *not* trust those models' absolute calibration. A simulated year proceeds in layers:

1. **Anchored ML prediction.** Each indicator's prediction is anchored to last year's value, and only the *policy signal* — the relative difference between the prediction under chosen levers and under baseline levers — is applied, capped at $\pm$`POLICY_SIGNAL_CAP` $= 3\%$. The models are used for the *relative* effect of policy only, because their absolute level is not trusted.
2. **Hand-written causal rules.** The causal graph (`sera.twin.causal_graph`) encodes documented lever→indicator links (e.g., healthcare spending → life expectancy, corporate tax → business density, education spending → completion rates and youth employment) with strength `CAUSAL_RULE_STRENGTH` $= 2\%$ maximum per lever per year. Lever values are normalized against the historical median and robust spread of the corresponding ISTAT series and squashed through tanh, so a "large" deviation is defined by history, not by an arbitrary scale. These rules exist because the statistical sensitivity of many indicators to policy in the historical data is weak: the rules add a guaranteed domain-knowledge elasticity so every documented link responds even where the data is too noisy for the models to learn it.
3. **Inter-indicator propagation (sign-aware, panel-reviewed).** Effects flow along indicator→indicator edges (income → poverty and inequality, unemployment → income and crime, business density → unemployment and wages, carbon emissions → air quality and health, ...) with a dampening factor, so a lever's first-order effect ripples through the system. Each edge now carries an explicit *direction*: the target moves with (+) or against (−) the source, taken from the panel-estimated coupling where the data supports it and from documented domain knowledge otherwise (Section 4.4). This corrects a previous

sign-*less* implementation that pushed every target in the direction of its source's change — mis-signing every opposite-polarity link such as income→poverty.

4. **Bounds and the realism cap.** Logical bounds are enforced (rates in $[0, 100]$, Gini in $[0, 1]$, life expectancy below 150, $\ldots$), and a speed limit caps the year-over-year change of any indicator at $\pm 6\%$. Without it, the multiplicative layers compound across the horizon and an optimizer can drive GDP to absurd values; at baseline levers, annual changes are far below the cap, so it constrains only aggressively-pushed scenarios.

An honest finding recorded in the model card: the trained models' relative response *saturates* the 3% cap for almost any lever deviation, so the cap value is the de-facto size of the ML layer's contribution, and the differentiated lever→indicator structure comes mostly from the hand-written rules. This is exactly the kind of statement a persuasive demo would omit; SERA treats stating it as part of the ethics of the system (Section 8.2). The constants are exposed as constructor parameters of the simulator precisely so that sensitivity analysis can re-run any scenario under weaker and stronger assumptions. That the ML layer's contribution turns out to be small, and that much of the causal core rests on hand-written rules, is acceptable here: the goal of SERA is not to deliver a forecast-grade digital twin of Italy — a task that would demand province-level policy data the historical panel simply does not contain (Section 4.4) — but to provide a concrete, inspectable system on which to demonstrate and experiment with the ethics of decision-support modelling. A twin that is approximate but *honest about where its structure comes from* serves that purpose better than a more elaborate model whose provenance is hidden; surfacing the limitation, rather than engineering it away, is itself the contribution.

4.4 What the data supports: estimating the twin's structure

A twin whose causal core is partly hand-written invites an obvious question: how much of that structure does the historical panel actually back? Rather than leave the answer to intuition, an analysis layer (`sera.twin.panel_estimation`) re-estimates the twin's relationships from the data as a *fixed-effects panel* — subtracting province and year means so that a rich province being persistently rich, or a national boom year, cannot masquerade as a learned relationship — and compares the result to what the twin asserts. Three findings, all reproducible via `tools/ad_hoc/estimate_twin_structure.py`:

- **The lever→indicator response is unidentifiable, by data availability.** All 20/20 policy levers are published only at the national level: there is no province-level corporate-tax rate or health-spending allocation to learn from, so the per-province lever values the trainer sees are population-share artifacts of disaggregation, mutually collinear and uninformative about policy. No model specification recovers a provincial policy elasticity that the data does not contain. This is the structural reason the hand-written rules, not the ML, must carry the lever response.

- **The indicator→indicator graph *is* mostly data-backed.** Re-estimated under fixed effects, the documented edge directions agree with the data on 76% of the 17 in-panel edges — well above chance, and a genuine endorsement of the domain graph. The fixed effects matter: the same comparison on the pooled specification the production trainer uses (no entity/year effects) scores only 59%, so the omitted contrast is part of why the pooled ML reads as noise.

- **Honest validation is much weaker than the in-sample fit.** The random train/test split the trainer reports gives a flattering $R^2 \approx +0.79$; a time-ordered holdout (train on the earlier years, test on the later ones) yields a strongly negative R^2 — worse than predicting the mean — and only 62% directional accuracy on next-year change. The optimistic number is a leakage artifact of shuffling adjacent province–years, and the model card now reports the temporal figure instead.

Acting on the second finding, the inter-indicator propagation is now *wired to the reviewed graph*: each edge's direction is the sign of its panel-estimated coupling where the edge is estimable, documented domain knowledge where it is not, and *dropped* where the data shows no link. Of the 56 hand-written edges, the review **flips 2** (whose documented sign the data contradicts — e.g. `school_enrollment`→`completion_rates`) and **drops 3** for lack of support (e.g. `unemployment_rate`→`crime_rate`, where `crime_rate` is a disaggregated series carrying little provincial signal). The magnitudes stay hand-tuned and bracketed by the sensitivity band; only the *directions* are data-governed, which is exactly what the analysis can defend. This is the one component the data lets the project move from asserted to estimated, and it does so without pretending the unidentifiable lever response can be learned too.

4.5 Policy levers and the budget constraint

The action space consists of 20 annual policy levers per province (Table 1). Levers are not free: the system applies a national budget constraint each simulated year. The available pool is a fixed share of national GDP scaled by the average *tax-lever* level relative to baseline,

$$\text{pool}_t \;=\; \beta \cdot \text{GDP}(s_t) \cdot \frac{1}{|T|} \sum_{\tau \in T} \frac{a_{\tau,t}}{a_\tau^{\text{base}}}, \tag{1}$$

where T is the set of four tax levers and β a spending-intensity share (≈19% of GDP, read from the public-finance data). Written at the national level for readability, the implementation actually sums each province's GDP-weighted tax ratio (`ui/backend_bridge.py`); the two coincide when the tax levers are uniform across provinces and differ only second-order otherwise. The thirteen spending levers consume the pool in proportion to their deviation above baseline; if total national cost exceeds the pool plus the carried-over reserve, all spending levers are scaled down toward baseline by the common feasibility ratio, and unspent budget carries over as a reserve (Algorithm 2). Cutting taxes below baseline therefore shrinks the budget available for programs — the fiscal trade-off that keeps "stimulate with tax cuts *and* maximize spending everywhere" infeasible. Effect sizes are calibrated so that no single lever family can push a province to the growth cap on its own: policies must make real choices, which is precisely what lets different ethical objectives reach genuinely different optima (Section 9).

Family	**Levers**
Taxation — fund the pool (4)	income tax, corporate tax, property/wealth tax, VAT
Budget allocations — spend the pool (8)	healthcare, education, infrastructure, social welfare, R&D incentives, green energy/environment, pensions, public-sector wages
Sector support — spend the pool (5)	agriculture, manufacturing, tourism, small business, housing/urban development
Regulatory / institutional — no budget cost (3)	immigration openness, regulatory burden, environmental-regulation strictness

Table 1: The 20 annual policy levers (`ANNUAL_PARAMETERS` in `causal_graph.py`). Four tax levers fund the national pool; the thirteen spending levers (budget allocations plus sector support, including public-sector wages as compensation spending — the `SPENDING_PARAMS` set in the budget code) consume it; the three regulatory levers carry no budget cost.

The budget is deliberately stylized — the model card lists what is absent (real fiscal multipliers, debt dynamics, EU fiscal rules) — but it is the ethically load-bearing component of the simulation: distributive justice is only a live question under scarcity, and the budget is what makes the simulated scarcity binding.

4.6 The rollout environment

A `RolloutEnv` wraps the simulator and rolls a policy forward over a multi-year horizon H: each year the policy maps the current provincial state to lever allocations, the budget constraint scales them to feasibility, the twin simulates the year, and the chosen ethical objective scores the resulting state. The optimization target is the cumulative welfare

$$J(\pi) \;=\; \sum_{t=1}^{H} W(s_t^{\pi})\,, \tag{2}$$

where s_t^{π} is the provincial state in year t under policy π and $W(\cdot)$ is the objective's per-year welfare function (Section 5). Scoring every year of the horizon rather than only the final year matters ethically: a policy that immiserates the early years to spike the final one is penalized, which is a (mild) form of intergenerational fairness within the horizon. The twin is a non-differentiable black box of sklearn models plus rules, so all trainable policies are gradient-free.

4.7 The control-room UI

The UI presents the system as a policy control room, and its layout encodes the oversight philosophy:

- **Header control panel.** The policy-model dropdown (with explainability badges) and the ethical-objective dropdown, each option carrying a plain-language description that names what the choice ignores ("a euro in Milan counts the same as a euro in Crotone"; "growth anywhere else counts for nothing"). Both choices must be made before optimization — the value judgment precedes the computation.
- **Province map and trends.** A choropleth of the 110 provinces; clicking a province opens its historical indicator trends, with history and simulation visually separated and simulated values labeled as model estimates.
- **Allocators and budget meter.** Per-province lever allocators and a national budget meter showing pool usage and the carry-over reserve, making the fiscal trade-off tangible while the user experiments manually.
- **Optimizer panel.** Returns the three graded candidates (full / moderate / baseline) with their efficiency and equity metrics and the sensitivity band (Section 8), plus the per-model explanation artifact — a weight matrix, a rule list, cluster tables, GP partial-dependence curves, or the neural audit.
- **Ethics equity dashboard.** The side-by-side comparison of all four frameworks (the experiment of Section 9 is exactly this view, run headlessly).
- **Pareto frontier view.** The NSGA-II front with the three named frameworks tagged as corners, every point expandable into its lever table.

5 Theoretical Analysis

This section analyzes the four implemented ethical frameworks: each is first discussed philosophically, then its algorithmic translation in SERA is presented and its design choices defended.

Throughout, $y_p \geq 0$ denotes the GDP per capita of province p in a simulated year, $n = 110$ the number of provinces.

A scoping note: in this domain the moral patients are *provinces*, not persons. Aggregate provincial indicators stand in for the wellbeing of residents, which imports a deliberate, documented simplification: within-province inequality (rich and poor households in Milan) is invisible to every objective. SERA names this value judgment in its ethics statement rather than hiding it in the code. A second scoping note: all four objectives are *patterned* principles in Nozick's sense — they evaluate end-state distributions, not the procedures that produced them; procedural and entitlement-based theories of justice have no natural encoding in an objective function over simulated states, which is itself an instructive limit of the optimization paradigm.

5.1 Utilitarianism — the philosophical view

Utilitarianism holds that the right action (or policy, or institution) is the one producing the greatest overall utility. Its roots reach back to classical hedonism — pleasure as the only intrinsic good, pain the only intrinsic evil — combined with a consequentialist standard of rightness, but it became a distinct school in the late 18^{th} century with Jeremy Bentham [1], who defined utility as the property of an object "whereby it tends to produce benefit, advantage, pleasure, or happiness, or to prevent the happening of mischief, pain, evil or unhappiness" and proposed a *felicific calculus* to measure it quantitatively along dimensions such as intensity, duration, certainty, and extent [3].

J.S. Mill [2] kept the maximizing consequentialism — the right action remains that which produces the greatest balance of pleasure over pain — but rejected Bentham's purely quantitative treatment, distinguishing *higher* from *lower* pleasures and holding that no quantity of a lower pleasure outweighs a significant quantity of a higher one: "it is better to be a human being dissatisfied than a pig satisfied" [4]. The distinction matters here because it anticipates the beyond-GDP critique taken up in Section 5.7: once utility has qualitative dimensions, a single monetary aggregate cannot represent it.

Contemporary taxonomies [3] distinguish *act*-utilitarianism (judge each act by its own consequences) from *indirect* forms: rule-utilitarianism (judge rules by the consequences of their general adoption) and institutional utilitarianism (the best political and social institutions are those producing the greatest total well-being). Since SERA evaluates standing *policies* adopted by an institution over a multi-year horizon, its utilitarian objective is best read as institutional utilitarianism with a GDP proxy for welfare: each marginal euro counts the same wherever it lands. The classic objection — utilitarianism permits sacrificing the few for the many, because persons are merely containers of aggregable utility — is not avoided; on the contrary, SERA is designed to make it *visible* when it happens (the equity dashboard's "who gains, who loses" maps, and empirically Figure 4).

5.2 Utilitarianism — the implementation

The utilitarian objective scores a simulated year by total national GDP,

$$W_{\text{util}}(s) \;=\; \sum_{p=1}^{n} y_p, \tag{3}$$

and the optimizer maximizes its cumulative sum (Eq. 2). Two properties matter ethically. First, the objective is distribution-blind by construction: a euro in Milan counts the same as a euro in Crotone, so the optimizer will happily concentrate levers where the modeled returns are highest. Second, the proxy itself embeds a value judgment that the documentation states explicitly: "national GDP" is the *unweighted* sum of provincial GDP per capita — small provinces weigh as much as large ones, which is itself a (mildly egalitarian) deviation from population-weighted utilitarianism. A population-weighted variant would be a one-line change; the point is that either choice is a choice.

5.3 Justice as Fairness — the philosophical view

Rawls' *Justice as Fairness* [5, 6, 7] positions itself against both utilitarianism and strict egalitarianism: it shares the former's concern for improving well-being but rejects sacrificing the worse-off to aggregate gains; it accepts the latter's premise of moral equality but permits inequalities that work to everyone's benefit. Rawls models society as "a fair system of social cooperation" among free and equal citizens, and derives its principles from the *Original Position*: a hypothetical choice situation behind a *Veil of Ignorance*, where the parties know general social facts but not their own place — class, talents, or conception of the good. Deprived of any basis for bias, Rawls argues, the rational choice is risk-averse: the parties evaluate candidate social structures by their worst possible position — the *maximin* rule — because gambling on a utilitarian structure could leave them in a sacrificed minority.

This reasoning yields two principles: first, each person has an equal claim to a fully adequate scheme of equal basic liberties; second, social and economic inequalities are permissible only if (a) attached to offices and positions open to all under fair equality of opportunity, and (b) arranged to the greatest benefit of the least-advantaged members of society — the *difference principle* [6].

The maximin derivation has been contested since Harsanyi [8] argued that rational parties behind the veil would maximize *expected* utility — recovering utilitarianism — rather than the minimum; the dispute is precisely over how much risk-aversion impartiality requires. SERA does not adjudicate it: it implements both endpoints and lets their consequences be compared on the same model (Section 9). The Veil of Ignorance has a natural reading in SERA's domain: a policymaker who did not know *which province they would live in* would, on Rawls' argument, want national policy to maximize the floor.

5.4 Rawlsian justice — the implementation

The Rawlsian objective is maximin over provinces:

$$W_{\text{rawls}}(s) \;=\; n \cdot \min_{p}\, y_p. \tag{4}$$

Only the worst-off province counts; the factor n merely rescales the score to "the national total Italy would have if every province lived like its poorest one," making the number comparable in magnitude to the utilitarian score in the UI's charts (the optimizers only ever compare candidates *within* one objective, so monotone rescaling is innocuous). Growth anywhere other than the minimum contributes nothing; the optimizer is thereby forced to direct lever budgets at the bottom of the distribution. Because the cumulative objective (Eq. 2) sums the minimum over every simulated year, the identity of the worst-off province may change as the floor rises — the optimizer chases whichever province is poorest *now*, a leapfrogging dynamic that is the difference principle in motion.

Two deliberate simplifications relative to Rawls are documented. First, the implementation applies the difference principle to a single primary good (provincial GDP per capita) rather than his index of primary goods. Second, it does not model the first principle (basic liberties) or fair equality of opportunity, which have no analogue in the lever space; what is implemented is the difference principle alone, which Rawls himself ranked lexically *after* the liberties it presupposes.

5.5 Egalitarianism — the philosophical view

Egalitarianism is less a single doctrine than a family of theories united by *moral equality* — the conviction that all persons have the same fundamental worth. As Bidadanure and Axelsen put it, the egalitarian task is not proving that we are equals but theorizing what it means to treat persons as such [9]. Historically this was a sharp break: classical thinkers treated hierarchy as natural, and moral equality emerged as a dominant ideal only in the 18th century — in Rousseau's claim that inequality is the product of society rather than nature, and in Kant's grounding of equal dignity in rational autonomy: persons have a dignity rather than a price, and must be treated always as ends in themselves.

Modern distributive egalitarianism, shaped in the 20th century by Rawls, Dworkin, Cohen, Sen, and Anderson, splits along two questions. The first is *what* should be equalized — the "equality of what?" debate [12]: resources, welfare, opportunity for welfare, or capabilities. The second is whether distribution is the right frame at all: *relational* egalitarians such as Anderson [11] argue that equality is a property of social relationships — the absence of domination and marginalization — not of allocations; a view that resists encoding in any objective function over indicator vectors, and is therefore out of scope here by the nature of the tool, a limit worth stating rather than ignoring.

Within distributive egalitarianism, the strictest variant — everyone gets the same level of goods [10] — faces the *leveling-down objection*: strict equality is satisfied by making everyone equally badly off, which seems to count as an improvement in at least one respect even when it benefits no one. Sen's welfare economics offers a measure that keeps equality morally weighty without rewarding leveling down: value total income *discounted* by its inequality, operationalized as mean income multiplied by one minus the Gini coefficient [13]. Growth still counts; unequally distributed growth counts less.

5.6 Egalitarianism — the implementation

SERA implements the Sen welfare function rather than strict equality of allocations, precisely to avoid rewarding leveling down:

$$W_{\text{egal}}(s) \;=\; \Big(\sum_{p=1}^{n} y_p\Big)\,\big(1 - G(y)\big), \qquad G(y) \;=\; \frac{2\sum_{i=1}^{n} i\, y_{(i)}}{n\sum_{i=1}^{n} y_{(i)}} - \frac{n+1}{n}, \tag{5}$$

where $y_{(1)} \leq \cdots \leq y_{(n)}$ are the sorted provincial values and G is the Gini coefficient in $[0,1]$, computed via the relative mean absolute difference. The objective is multiplicative: a policy that raises total GDP by 5% while widening the North–South gap can lose to one that raises it 3% evenly. An equal distribution gives $G = 0$ and recovers the utilitarian score; a maximally concentrated one is discounted toward zero. The inequality that is measured — and the only one visible to the objective — is *inter-provincial*; this is restated wherever the objective is described.

5.7 Beyond GDP: multi-dimensional wellbeing — the philosophical view

A separate critique cuts across all three frameworks above: whatever the distribution rule, *GDP is the wrong currency.* Sen's capability approach [12] argues that resources are means, not ends; what matters is what people can do and be — their capabilities — of which income is an imperfect and sometimes misleading proxy. The same conviction drove the Stiglitz–Sen–Fitoussi Commission's recommendation that policy evaluation move from production measures to multi-dimensional wellbeing dashboards [17]: health, education, employment, environment, and material conditions alongside income. Italy itself institutionalized this view in ISTAT's BES (*Benessere Equo e Sostenibile*) framework, which feeds wellbeing indicators into the national budget process — making a multi-indicator objective not an academic exercise but a description of how the data provider itself believes its data should be used. On this view the prior objectives do not just weigh provinces wrongly; they measure the wrong thing about each province.

5.8 Multi-objective wellbeing — the implementation

The wellbeing objective scores a year by a weighted composite of percent changes from the simulation's starting state:

$$W_{\mathrm{wb}}(s) \;=\; 100\,\frac{\sum_{k\in K}\omega_k\,d_k\left(\frac{\bar{x}_k(s)}{\bar{x}_k(s_0)}-1\right)}{\sum_{k\in K}\omega_k}, \tag{6}$$

over the components k *present in the panel* $K \subseteq$ {GDP per capita ($\omega = 0.35$, $d = +1$), life expectancy (0.25, +1), unemployment rate (0.20, −1), poverty rate (0.20, −1)}, where $\bar{x}_k$ is the national mean of indicator k, s_0 the starting state, and d_k the desired direction. The denominator renormalizes the weights over whatever components are available (in the experiments all four are present, so it is 1 and the weights are used as written); the code carries it so a missing indicator reweights the rest rather than silently shrinking the score. Anchoring to the starting year makes the score read as "composite improvement in percent points," commensurable across indicators with wildly different units. The weights are, unavoidably, value judgments; SERA's position is not that these weights are right, but that they are *written down* where the three other objectives can be selected instead. The composite ignores everything not in it — stated in the ethics table verbatim. Note also that this objective aggregates national means, so it is distribution-blind *across* provinces in each component: it changes the currency of welfare, not the distribution rule. Currency and pattern are orthogonal axes, and SERA's four objectives sample three patterns of one currency plus one alternative currency.

5.9 Three further objectives: a smoothed maximin and two parameterized families

The four objectives above are the analytical backbone — three distribution rules over one currency plus one alternative currency. The registry is deliberately extensible, and three further objectives are implemented to round out the distributive-justice space and to support the experiments of Section 9. The two that an earlier version of this project left out on an "arbitrary-constant honesty" argument are now included precisely by turning that constant into an explicit, user-facing parameter — which is more in the spirit of the project than omission was.

Smoothed maximin (CVaR). Hard maximin (`rawlsian`) scores a single province out of 110, the sparsest training signal in the registry. The `cvar` objective keeps the Rawlsian spirit — it cares only about the bottom of the distribution — but averages the worst α fraction of provinces (a Conditional-Value-at-Risk reading of the difference principle):

$$W_{\mathrm{cvar}}(s) \;=\; n\cdot\frac{1}{|L_\alpha|}\sum_{p\in L_\alpha} y_p, \qquad L_\alpha = \text{the } \lceil \alpha n\rceil \text{ poorest provinces.} \tag{7}$$

As $\alpha \to 0$ it recovers pure maximin; at $\alpha = 1$ it is the (rescaled) utilitarian mean. Its purpose is experimental: comparing it against hard maximin on the *same* budget separates "maximin the theory is destructive" from "maximin the objective is hard to optimize" (Section 9).

Sufficientarianism. Frankfurt [14] argues that "equality is not, as such, of particular moral importance": what matters morally is not that everyone has the same but that each has *enough*; inequalities above the sufficiency threshold carry no moral weight. Timmer's modern formalization [15] structures the view as priority ranges on a continuum with one or more thresholds. The `sufficientarian` objective scores a year by the negative total shortfall below a threshold θ, $W_{\mathrm{suff}}(s) = -\sum_p \max(0, \theta - y_p)$. Because *the threshold* θ *is the entire theory* and there is no non-arbitrary euro value for "a sufficient province," θ is not hard-coded: it is set *relative to the* *starting distribution* (a fraction of the initial median provincial GDP) and exposed as a user-facing slider, so the value judgment is made visibly by a person rather than smuggled in as a default — exactly the move the project exists to force.

Prioritarianism. Parfit [16] proposes that benefits matter more the worse off their recipient is — a strictly concave weighting $W_{\text{prio}}(s) = \sum_p u(y_p)$ with $u' > 0$, $u'' < 0$ — giving priority to the worse-off without Rawls' exclusive focus on the minimum and without egalitarianism's intrinsic concern for relativities (it is a view about *levels*, not gaps, so it evades the leveling-down objection entirely). The `prioritarian` objective implements a CRRA-style $u(y) = y^{1-\rho}$ whose concavity $\rho \in [0, 1)$ is a user-facing slider: at $\rho = 0$ it is exactly utilitarian, and as $\rho \to 1$ the optimizer's behavior slides toward maximin, so the single parameter sweeps the whole continuum between SERA's endpoints. This makes the continuum the Pareto frontier was meant to expose (Section 5.10) selectable as well as inspectable; and because the experiment of Section 9 shows that over *uniform* national levers that continuum collapses, the prioritarian objective is most interesting paired with the per-province and clustered policies that can actually act on it.

5.10 From a dropdown to a frontier: the NSGA-II Pareto search

Each objective above forces the efficiency–equity trade-off into a single scalar *before* optimization. The Pareto module (`sera.twin.pareto`) drops that framing: an NSGA-II [23] evolutionary search evolves uniform national lever vectors against three objectives *simultaneously* — total final-year GDP (efficiency), inter-provincial Gini (inequality, minimized), and the worst-off province's GDP (the floor) — and returns the non-dominated front (Algorithm 5). A point dominates another if it is at least as good on all three criteria and strictly better on one; the front is the set of policies no other policy dominates. The UI plots the frontier with the utilitarian, egalitarian, and Rawlsian optima tagged as *corners of the same curve*, and every point's lever vector remains inspectable as a table.

The ethical significance is the reframing itself: the trade-off becomes a continuum to be inspected rather than a dropdown to be trusted. A decision-maker sees not "the AI's answer" but the shape of what is jointly achievable — how much total GDP must be conceded to raise the floor by a given amount. The search space (one shared lever vector in $[0, 1]^{20}$) keeps the population search affordable against the slow twin and keeps every candidate white-box. Like the equity dashboard, the frontier is a read-only what-if: it never advances the twin. Section 9 maps it empirically.

6 The Policy-Model Spectrum

The second user-facing choice is *how* to search. All trainable policies are objective-agnostic — the framework being maximized comes entirely from the rollout environment — and gradient-free. They differ in their *policy class*: how expressive the mapping from provincial state to levers is, and how auditable the trained result remains. This section details each model; Section 7 discusses why the spectrum exists.

6.1 Baseline

`BaselinePolicy` holds every lever at its historical baseline — the reference scenario against which every intervention is measured, and always one of the three candidates presented to the user.

6.2 Neural policy (black box)

`NeuralPolicy` is a small pure-NumPy MLP applied *per province*. For province p at horizon step t, the feature vector is

$$z_p \;=\; \Big[1,\; t/H,\; \operatorname{clip}\big(x_{p,k}/\bar{x}_k - 1,\, \pm 3\big)_{k\in F}\Big], \tag{8}$$

where F is a six-indicator feature set (GDP per capita, unemployment, income, life expectancy, business density, poverty rate) and $\bar{x}_k$ the national reference mean locked in from the starting state — so each province's *deviation from the national profile* is what the network reads. The levers are

$$a_p \;=\; a^{\min} + \sigma\big(W_2 \tanh(W_1 z_p + b_1) + b_2\big) \odot (a^{\max} - a^{\min}), \tag{9}$$

with 16 hidden units and σ the logistic function, giving every province its own tailored lever vector from one shared weight set. The weights are trained with mirrored-sampling Evolution Strategies [24] (Algorithm 3): perturbations $\pm\varepsilon_i$ are evaluated by full rollouts, advantages are normalized, and the weighted perturbation sum estimates an ascent direction on $J(\pi)$. It is the most expressive model in the registry and the least readable — hence the post-hoc audit machinery of Section 7.

6.3 Linear policy (white box)

`LinearPolicy` is the identical per-province setup with the hidden layer removed: $a_p = a^{\min} + \sigma(W z_p) \odot (a^{\max} - a^{\min})$. The signed weight matrix W *is* the explanation: a positive weight means the lever rises when that indicator is above the national reference, and every lever's response to every indicator is one readable, contestable number. Crucially it trains with the same ES loop, hyperparameters, and rollout budget as the neural policy, so any score gap between them is a measured price of transparency rather than an artifact of a different optimizer.

6.4 Rule list (white box)

`DecisionListPolicy` gives each lever one human-readable threshold rule on one indicator's deviation from the national reference:

$$\text{IF } x_{p,k} > \bar{x}_k(1+\tau) \text{ THEN } a_{p,\ell} = A \text{ ELSE } a_{p,\ell} = B, \tag{10}$$

where the rule's indicator k (via selection logits), threshold $\tau \in [-25\%, +25\%]$, and levels A, B are all part of one genome searched with the Cross-Entropy Method (Algorithm 4). The trained policy prints as a short list of sentences a non-technical reader can audit — the most transparent per-province model in the registry.

6.5 Regional clusters (white box)

`ClusterLeverPolicy` sits between one national vector and 110 implicit ones: provinces are grouped by their starting indicators with a deterministic k-means (default $k = 4$), and CEM searches one shared lever vector per cluster — the way real governments write "plans for the industrial north" or "plans for the Mezzogiorno." The explanation is the cluster membership plus one small lever table per cluster.

6.6 Uniform national policy — CEM and Bayesian (white/gray box)

`UniformLeverPolicy` applies the *same* lever vector to every province, reducing the search space to one point in $[0, 1]^{20}$, found with CEM: a diagonal Gaussian over the lever box is iteratively refit to the elite fraction of sampled candidates, with a variance floor to keep exploring. The result is one readable national lever table.

`BayesianUniformPolicy` finds the same vector with Gaussian-process Bayesian optimization [26]: because every twin rollout is expensive (110 provinces $\times$ H years), a GP surrogate of $J(\cdot)$ over the lever box is fit to the evaluations so far and each next rollout is chosen by Expected Improvement — typically an order of magnitude fewer rollouts than population search. The fitted surrogate doubles as the explanation: per-lever partial-dependence curves *with the GP's own uncertainty*, i.e., "how sure is the optimizer that raising this lever helps, under this ethical objective." It is labeled a gray box: more legible than the neural policy, less direct than a weight matrix.

6.7 Blended interventions

`BlendedPolicy` wraps any policy and scales its levers toward baseline: $a^{\lambda} = a^{\text{base}} + \lambda\,(a - a^{\text{base}})$ with $\lambda \in [0, 1]$. It implements the "moderate intervention" candidate ($\lambda = 0.5$) in the oversight workflow — graded options instead of a single optimum.

7 Explainability as an Ethical Design Axis

A policy that allocates public money should be *contestable*: someone affected by it must be able to see why their province got the levers it got. SERA treats explainability not as an afterthought but as a dimension of the model space itself, in dialogue with Rudin's argument that high-stakes decisions should prefer inherently interpretable models over post-hoc explanations [20].

Model	Search	Badge	What can be audited
`baseline`	none	white box	historical levers
`rules`	CEM	white box	one IF/THEN threshold rule per lever; the whole policy prints as sentences
`uniform_cem`	CEM	white box	a single national lever table
`cluster_cem`	k-means + CEM	white box	cluster membership plus one lever table per cluster (regional packages)
`linear`	Evolution Strategies	white box	the full signed weight matrix: every lever's response to every indicator
`uniform_bayes`	GP Bayesian opt.	gray box	per-lever partial-dependence curves with the surrogate's own uncertainty
`neural`	Evolution Strategies	black box + audit	permutation importance; distilled surrogate tree with an honest fidelity score

Table 2: The policy-model explainability spectrum. Every model carries a badge in the UI and produces an explanation artifact next to its candidates.

Three commitments shape the design:

1. **The transparent models compete on equal terms.** The linear policy is the identical per-province setup as the neural one with the hidden layer removed, trained with the *same* mirrored-sampling Evolution Strategies loop [24]. Any score gap between them is therefore a *measured* price of transparency, visible per run — not an assumption in either direction. This operationalizes the empirical half of Rudin's claim: whether black boxes actually buy performance is a question the system answers with data, per problem instance.

2. **Post-hoc explanations state their own limits.** The neural policy is audited (`sera.twin.explain`) with two instruments. The first is *permutation importance*: one indicator is shuffled across provinces and the mean absolute shift of the policy's lever choices (in lever-range units) is measured — answering "which indicators is this policy actually reacting to?". The second is *surrogate distillation*: a shallow decision tree is fit to

imitate the network's decisions, and its *fidelity* — held-out R^2 against the network's own decisions — is reported, not assumed. The tree's leaves read as province segments with lever packages; a segment list with fidelity 0.6 is presented as a sketch, not as the policy. Both instruments operate on the policy's decisions only, with no extra twin rollouts, so they are cheap enough to run after every optimization.

3. **Explaining the policy is not validating the twin.** Every artifact describes how a policy model maps indicators to levers *inside a simulation* whose causal core is hand-written. Explainability here serves contestability, not proof of correctness.

The deeper point is that explainability interacts with the ethical objective. A Rawlsian policy that cannot be explained invites the suspicion that it is not actually Rawlsian — that the optimizer found a degenerate exploit rather than a principled redistribution. White-box policies make the implemented ethics *verifiable*: one can read the rule list and check that it does, in fact, route spending toward lagging provinces. For the black-box policy, the audit artifacts are the next best thing: importance scores showing which indicators drive its decisions, and a surrogate whose own honesty metric is displayed beside it.

8 Human Oversight and Honesty about Uncertainty

8.1 Candidates, not commands

The optimizer never applies anything automatically. Each run returns three graded candidates — *full intervention* ($\lambda = 1$), *moderate* ($\lambda = 0.5$, via `BlendedPolicy`), and the historical *baseline* ($\lambda = 0$) — each with its efficiency/equity trade-offs (total GDP, inter-provincial Gini, worst-off province, unspent budget). A human must explicitly adopt one, including, as a first-class option, the baseline "do nothing." Adoption is explicit, single-shot, and recorded in the UI; fresh candidates must then be generated from the new state. The objective dropdown forces the value judgment to be made *before* optimization, by a person. The ethics statement makes removing any of this — the candidate step, the sensitivity band, the baseline option — an explicitly prohibited use: collapsing the workflow into a single "the AI says do X" answer is treated as a misuse of the system, not a UX simplification.

8.2 Honesty about uncertainty

The twin's lever→indicator response comes in significant part from hand-written causal rules layered on statistically weak signals. SERA's response to that fact is procedural:

- **Sensitivity bands.** Every optimizer run re-simulates the proposed policy with the causal rules at 0.5× and 1.5× strength (the simulator exposes `causal_rule_strength` as a parameter for exactly this purpose) and shows the spread as a shaded band. A wide band means the projection is mostly assumption. The band is a sensitivity analysis, not a confidence interval, and the UI says so.

- **No forecast claims — now measured, not just asserted.** The indicator models are used only for relative policy response, and a temporal backtest confirms why that restraint is necessary: a time-ordered holdout gives a strongly negative R^2 (worse than predicting the mean) and only 62% directional accuracy, against a flattering random-split $R^2 \approx +0.79$ (Section 4.4). Any claim of real-world accuracy is unsupported; outputs are internally consistent scenarios, not predictions.

- **Structural limits stated.** The ±6%/year cap excludes shocks and regime changes; the budget is stylized; the active 24-indicator panel carries no demographic variables (population, birth rate, and migration are not among the simulated indicators), so there is no demographic feedback in these experiments — the immigration and housing levers do wire to those indicators in the causal graph, but those edges stay inert until the demographic series are added to the panel; optimizers exploit the *simulator*, and transferring a high-scoring policy to Italy would be a category error.

8.3 Documentation as an ethical deliverable

SERA ships the documentation a responsible AI system needs: an ethics statement (`ETHICS.md`) covering intended use, prohibited uses, embedded value judgments, and affected groups; a model card [21] for the twin and policy models including honest validation status; and a datasheet [22] for the ISTAT dataset including who is missing from the data. Affected groups are named explicitly: people invisible in official statistics; residents of "unprofitable" provinces under the utilitarian objective (the reason the other objectives and the equity dashboard exist); and decision-makers themselves, via automation bias — a polished dashboard invites more trust than a hand-tuned simulation deserves. The uncertainty banner, the sensitivity band, and the mandatory candidate choice are described in the ethics statement as *mitigations, not solutions* to that bias.

8.4 EU AI Act positioning

As distributed, SERA is a research/educational prototype, not a deployed AI system. However, were a system like it used by a public authority to inform allocation of public funds or essential services, it would plausibly fall in the **high-risk** category of the EU AI Act (Regulation (EU) 2024/1689, Annex III) [27]. The design anticipates the corresponding obligations deliberately (Table 3); the ethics statement insists this mapping is a starting checklist for anyone moving SERA toward real use, not compliance.

AI Act obligation (Art. 9–15)	Where SERA addresses it
Risk management & known limitations	ethics statement; model card "Limitations"
Data governance	datasheet (provenance, gaps, preprocessing)
Technical documentation	README, model card, datasheet
Transparency to users	in-UI uncertainty banner; objective descriptions; explainability badges and per-run explanation artifacts
Human oversight (Art. 14)	candidate/adopt flow; baseline as first-class choice
Accuracy & robustness	sensitivity band; realism caps; documented validation gaps

Table 3: EU AI Act high-risk obligations mapped to SERA's design.

9 Experimental Comparison of the Ethical Frameworks

This section reports the experiment the system was built to enable: the same policy model, trained once per ethical framework from the same starting state, under the same budget, on the same twin — so that the only variable is the moral criterion. It is the headless equivalent of the UI's ethics equity dashboard, followed by the NSGA-II frontier mapping. Everything is a read-only what-if: the twin state is never advanced.

9.1 Experimental setup

- **Starting state:** the latest historical values of the 24-indicator panel for all 110 provinces, anchored at 2025 (indicators missing 2025 fall back to their most recent available year, as logged by the loader; see Section 4.2).

- **Policy model:** `neural` — the per-province neural policy, chosen because it is the most expressive model in the registry and therefore gives each framework the greatest room to express its distributional preferences across provinces.
- **Training:** 6 iterations of mirrored-sampling ES (the UI's default), population 6 (12 rollouts per iteration) — identical search budget across frameworks. To separate genuine framework effects from gradient-free search luck, every framework is trained under 6 independent seeds and all final-year metrics below are reported as *mean ± standard deviation across seeds.*
- **Frameworks:** the four canonical objectives plus the smoothed-maximin `cvar` ($\alpha = 0.2$), included specifically to test whether hard maximin's poor floor is the theory or its one-province training signal (Section 5.9).
- **Horizon:** 10 simulated years (2026–2035), cumulative-welfare objective (Eq. 2), national budget constraint active with reserve carry-over.
- **Baseline:** the historical-lever policy rolled out over the same horizon.
- **Pareto search:** NSGA-II, population 12, 6 generations (84 twin rollouts), seed 0 — run twice, once over uniform national lever vectors and once over per-cluster vectors (4 k-means regional packages) to test whether targeting recovers a trade-off uniform policy cannot express (Section 9.6).
- **Continuum sweep and ablation:** the prioritarian objective is additionally trained across a ladder of concavities (Section 9.7), and the headline comparison is re-run under three inter-indicator propagation rules to measure how much the data-reviewed edge signs matter (Section 9.9).

All numbers below are produced by `report/run_experiments.py`, `run_prioritarian_sweep.py`, `run_ablation.py`, and `report/plot_results.py`; no result is hand-typed. The usual caveat applies with full force: these are properties of the *simulator*, not of Italy (Section 8.2).

9.2 Headline outcomes

Table 4 reports the final-year (2035) outcomes of each framework against the baseline scenario on the three evaluation axes: total national GDP (efficiency), inter-provincial Gini on GDP per capita (inequality), and the worst-off province's GDP per capita (the floor). Figure 2 shows the full trajectories.

Objective	Total GDP ($\times 10^3$)	vs. base	Gini	Worst prov.	vs. base
Baseline (no intervention)	274.6	—	0.439	230.1	—
Utilitarian (total GDP)	305.1	+11.1%	0.437	250.1	+8.7%
Rawlsian (maximin)	251.1	-8.6%	0.377	271.0	+17.8%
Rawlsian (CVaR, worst 20%)	263.1	-4.2%	0.370	272.8	+18.5%
Egalitarian (Sen welfare)	298.7	+8.8%	0.406	274.6	+19.3%
Wellbeing (composite)	301.9	+9.9%	0.437	253.1	+10.0%

Table 4: Final-year (2035) outcomes of the same neural policy trained under each ethical framework from the same 2025 state, as the mean over 6 replication seeds (per-seed spread is reported in the text and Figure 2's bands). "Total GDP" is the sum of provincial GDP per capita (the system's national proxy); "worst prov." is the lowest provincial GDP per capita. Values are in the twin's native model units: disaggregated indicator levels are population-share estimates (Section 4.2), so absolute levels are not euro-interpretable and the meaningful comparisons are relative.

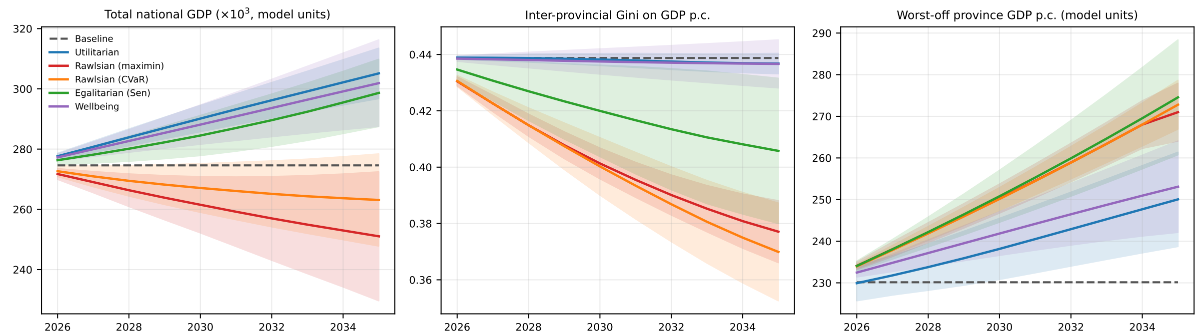

Figure 2: Simulated trajectories 2026–2035 under each ethical framework (same neural policy, same training budget, same budget constraint) against the baseline scenario: total national GDP (left), inter-provincial Gini on GDP per capita (center), worst-off province's GDP per capita (right). Lines are the mean over 6 seeds; shaded bands are ±1 standard deviation across seeds, i.e. the run-to-run spread of the gradient-free search.

Training improved the cumulative objective score over the untrained network by +23.5% (utilitarian), +27.5% (Rawlsian), +27.3% (CVaR), +25.5% (egalitarian), and +72.2% (wellbeing), averaged over the 6 seeds within the six-iteration budget — the per-run training quality report that the model card requires every optimization to disclose. Figure 3 plots the full convergence curves: how much of each framework's improvement is already banked by iteration 6, and how the sparsest objective (hard maximin) climbs slowest.

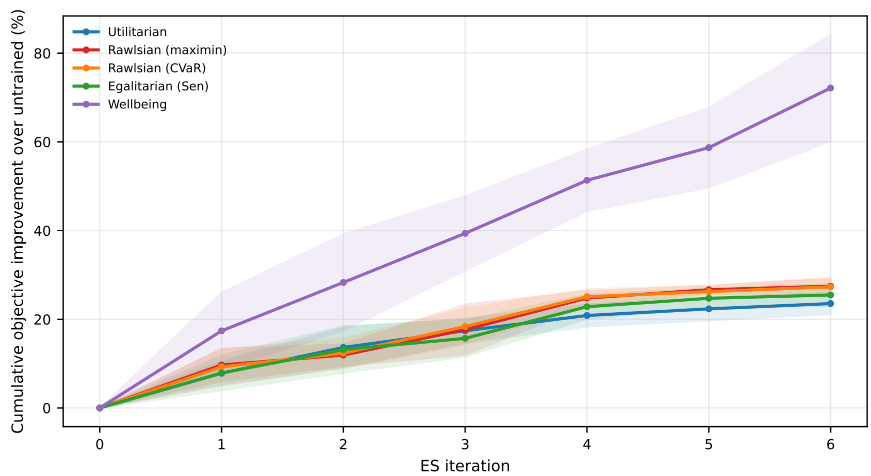

Figure 3: Training convergence per framework: cumulative objective improvement over the untrained network (%) versus ES iteration, mean over 6 seeds (bands ±1 std). Plotted as relative improvement so the four very different objective scales are comparable on one axis. Hard maximin's one-province signal gives it the least to climb on; the smoothed CVaR variant, scoring the worst 20% of provinces, gets a denser gradient.

9.3 Who gains, who loses

Aggregates hide the distributional signature of each framework, which is the ethically relevant part. Figure 4 disaggregates the final year to provinces: for each framework, the distribution over the 110 provinces of the percent change in final-year GDP per capita relative to the baseline scenario — the headless counterpart of the UI's "who gains, who loses" maps.

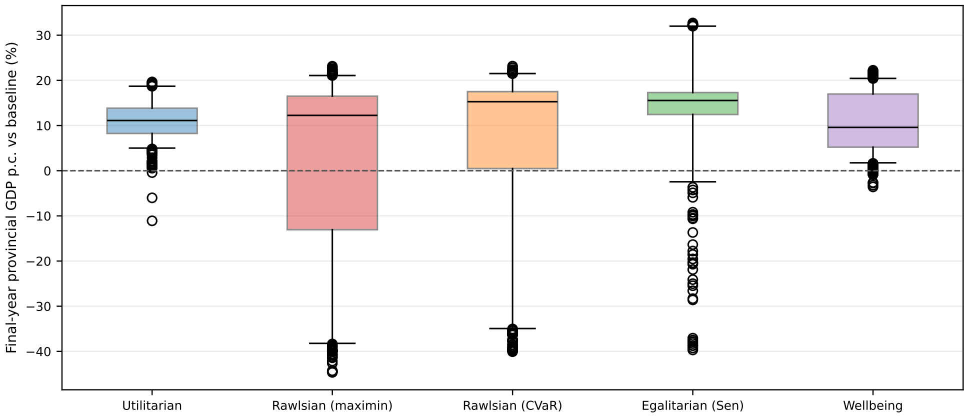

Figure 4: Distribution over the 110 provinces of final-year GDP per capita change versus the baseline scenario, per ethical framework (boxes: quartiles; whiskers: 5th–95th percentile). The frameworks differ not only in their means but in the *shape* of who benefits.

9.4 The North–South divide

The introduction chose Italy because of one cleavage in particular — the North–South divide — yet the results so far speak only in province-agnostic aggregates. This section reads them in the dimension that motivated the project. The 110 provinces are grouped into Italy's standard statistical macro-areas (`province_mapping.PROVINCE_TO_MACROAREA`): the 47-province North (Nord-Ovest + Nord-Est), the 22-province Centre, and the 41-province South, or *Mezzogiorno* (Sud + Isole). In the starting state the South's mean provincial GDP per capita is 2257 against the North's 2783— a South/North ratio of 0.811, the divide the literature describes. Table 5 and Figure 5 report what each framework does to it.

Objective	North Δ	South Δ	South/North ratio
Baseline	—	—	0.811
Utilitarian (total GDP)	+10.7%	+11.2%	0.815
Rawlsian (maximin)	-13.0%	-2.3%	0.913
Rawlsian (CVaR)	-8.9%	+2.1%	0.912
Egalitarian (Sen)	+6.7%	+12.2%	0.855
Wellbeing	+9.9%	+9.9%	0.811

Table 5: North vs South response by ethical framework: mean provincial GDP per capita change from baseline for the North and the Mezzogiorno, and the resulting South/North ratio (higher = a narrower divide; baseline 0.811). Means over 6 seeds.

Four territorial signatures emerge, and they sharpen the abstract results:

- **Utilitarianism and wellbeing leave the divide untouched.** Both lift North and South by almost exactly the same proportion (+10.7%/+11.2% and +9.9%/+9.9%), so the South/North ratio barely moves from baseline (0.815, 0.811vs. 0.811). A distribution-blind rising tide raises the Mezzogiorno in step with the North — it neither closes nor widens the gap, it *photocopies* it onto a richer Italy.
- **Maximin closes the divide by leveling the North down.** It produces the most equal territorial outcome of any framework (ratio 0.913) — but the mechanism is the leveling-down objection in geographic form: the North *falls* -13.0% while the South barely moves (-2.3%). The gap narrows because the top is cut, not because the bottom rises.
- **CVaR closes it more humanely.** The smoothed maximin reaches essentially the same

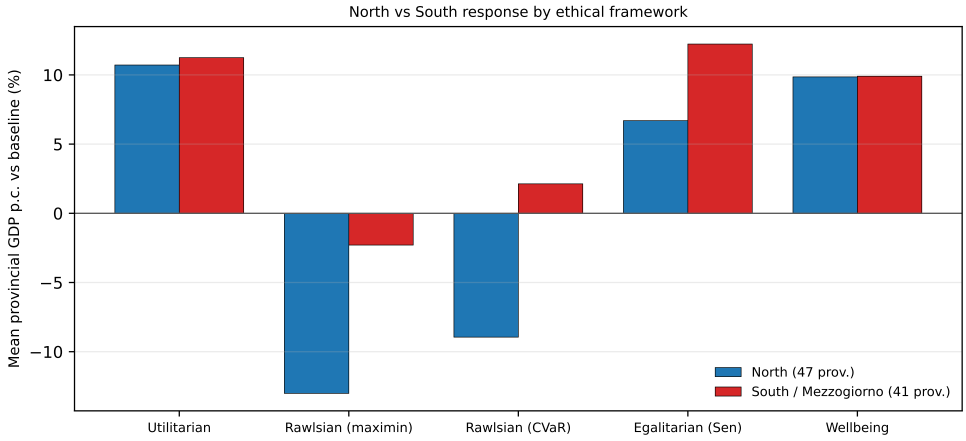

Figure 5: North (blue) vs South/Mezzogiorno (red) mean provincial GDP per capita change from baseline, per framework (6-seed means). A taller red bar than blue means the framework closes the divide by lifting the South; a deep blue trough means it closes the divide by pulling the North down.

ratio (0.912) but the South actually *gains* (+2.1%) while the North falls less (-8.9%) — a strictly kinder route to the same territorial equality, and the Mezzogiorno's only Rawlsian-family improvement.

- **Sen welfare is the cohesion result.** Egalitarianism is the one framework that narrows the divide (ratio 0.855) *while both regions grow*: it lifts the South faster than the North (+12.2% vs. +6.7%), the textbook definition of territorial cohesion — convergence without sacrifice.

Two honesty notes. First, the *absolute* North–South gap in the twin is a property of the disaggregated starting data (GDP is one of the regionally disaggregated indicators; Section 4.2), so the numbers above are read as each framework's *change* to the gap within the simulator, not as a measurement of Italy's real divide. Second, precisely because of that, the macro-area average over 47–41 provinces is a *sounder* unit than the single worst-off province the headline metrics lean on: aggregating washes out the per-province disaggregation noise, so the territorial reading here is, if anything, more robust than the floor it complements.

9.5 The efficiency–equity frontier

Figure 6 shows the NSGA-II result: 8 non-dominated uniform national policies from 84 twin rollouts. Across the front, total GDP spans 233.7–321.0 ($\times 10^3$) and the worst-off province's GDP per capita spans 195.9–269.0 — and the two move together with correlation 1.000, while the inter-provincial Gini is invariant across the entire front to within $\sim 7\times 10^{-6}$ (at 0.439 everywhere, to three decimals).

This degeneracy is the experiment's most instructive structural finding: *in the space of uniform national policies, there is no efficiency–equity trade-off to map.* A single lever vector applied identically to every province moves all provinces nearly proportionally in the twin's multiplicative dynamics — a stronger national stance produces more total GDP *and* a higher floor in lockstep, and relative dispersion barely responds. The expected three-cornered frontier collapses to a ray, and the tagged "corners" pile onto its ends (the utilitarian and Rawlsian corners coincide at the high end; the egalitarian tag lands on a point distinguished only by Gini noise at the 10^{-6} scale). The genuine distributional divergence seen in Table 4 — Gini moving from 0.439 to 0.377 — was produced by the *per-province* neural policy, which can treat provinces differently. The conclusion writes itself: in this twin, equity is purchased by *targeting*, not by the overall level of the national policy stance; an ethical framework only has distributional teeth if the policy class can discriminate between its moral patients.

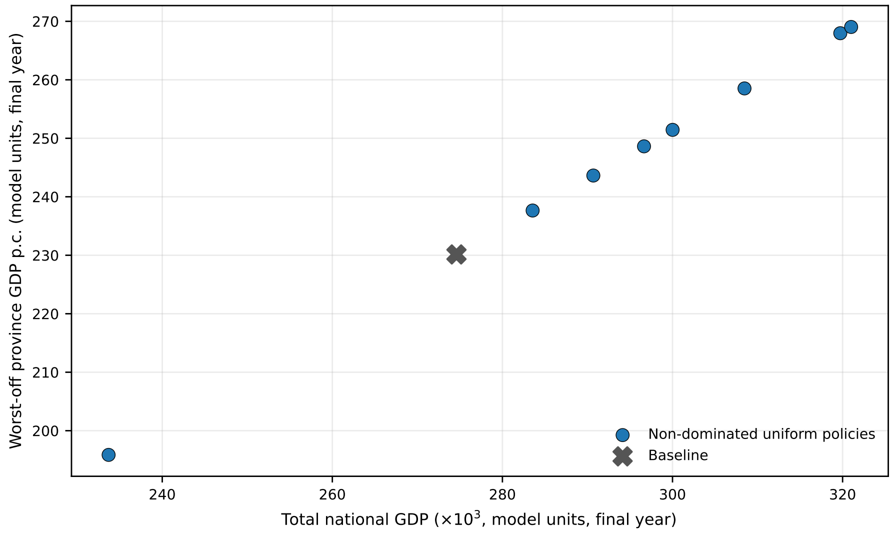

Figure 6: The NSGA-II non-dominated set over uniform national lever vectors (6 generations, population 12; final-year metrics), plotted as efficiency versus the floor. The set degenerates to a ray: total GDP and the worst-off province co-move (correlation 1.000) and the Gini is invariant to $\sim 10^{-6}$, so uniform national policy cannot trade efficiency against equity — only per-province targeting can (Table 4). The gray cross marks the baseline scenario.

9.6 The frontier returns once policy can target: the clustered search

The uniform degeneracy predicts its own remedy: if equity is purchased by targeting, then giving the Pareto search the *ability* to target should make inequality move. Re-running the identical NSGA-II over per-cluster lever vectors — 4 k-means regional packages instead of one national vector, the same population and generation budget — does exactly that (Figure 7). The front is small (3 non-dominated policies; an 80-dimensional genome is under-resolved by 84 rollouts), but it already does what the uniform front could not: the inter-provincial Gini *varies*, from 0.411 to 0.418 across the front, against the uniform front's invariance to 10^{-6}. The mechanism is exactly the predicted one — the lowest-Gini policy holds the richest cluster back while the highest-GDP policy feeds it. One nuance the numbers insist on: total GDP and the *floor* still co-move along the clustered front (they rise together from the egalitarian to the utilitarian corner), so what targeting opens is an efficiency-versus-*inequality* trade-off, not an efficiency-versus-floor one. That is the same lesson as Table 4, now isolated in a white-box, read-only search: *inequality responds to who is treated differently, not to how hard the single national lever is pushed.* The regional packages remain fully inspectable — each front point expands into one lever table per cluster, the way a real "plan for the Mezzogiorno" would read.

9.7 The prioritarian continuum, swept

The objectives so far are discrete points; the prioritarian objective (Section 5.9) is instead a *dial.* Its concavity ρ interpolates between utilitarian (ρ=0, identical by construction) and maximin ($\rho \to 1$), so sweeping it should trace the efficiency–floor continuum that the uniform Pareto search could not. Training the neural policy at a ladder of concavities (3-seed means; Figure 8) confirms it: as ρ rises from 0 to 0.95, total GDP moves from +11.2% to +8.1% versus baseline while the worst-off province rises from +10.3% to +17.1%, and the Gini falls from 0.435 to 0.407. The ρ=0 end reproduces the utilitarian objective by construction; its sweep figures differ from the

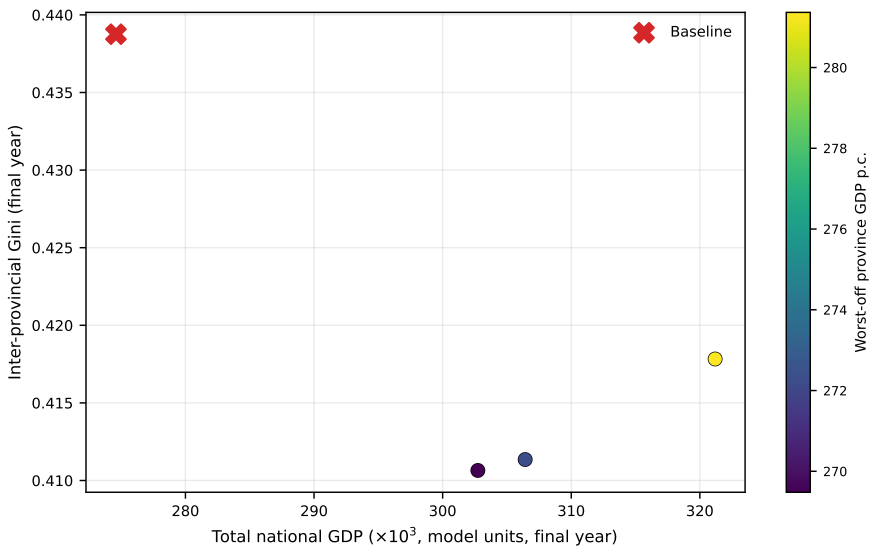

Figure 7: NSGA-II over *per-cluster* lever vectors (4 k-means regional packages; 6 generations, population 12). Unlike the uniform front, this one spans a real range of inequality (Gini 0.411–0.418), so efficiency (horizontal) and equity (vertical) are genuinely in tension; colour encodes the worst-off province's GDP. The red cross marks the baseline. Targeting — not the overall intensity of national policy — is what opens the trade-off.

headline utilitarian row of Table 4 only by seed noise (this sweep averages 3 seeds against the headline's 6, and the floor is the highest-variance axis). A single scalar slides the optimizer from the rising tide to the floor-first regime — the prioritarian gradient that sits, philosophically and now empirically, between SERA's two endpoints, and a concrete way for a decision-maker to choose *how much* priority the worst-off get rather than toggling between extremes.

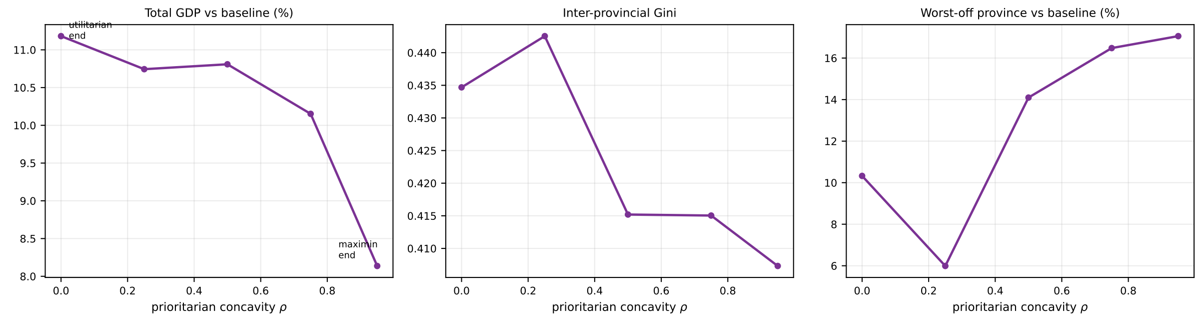

Figure 8: Prioritarian sweep: final-year total GDP (left), inter-provincial Gini (center), and the worst-off province (right) as the concavity ρ rises from the utilitarian end (ρ=0) toward maximin. Means over 3 seeds. The single parameter traces the efficiency–equity continuum as a curve.

9.8 The sufficiency threshold, swept

The other tunable objective is sufficientarian: minimise the shortfall below a threshold θ, where everything above θ is morally irrelevant and θ *is* the theory (Section 5.9). Sweeping it (3-seed means, θ as a multiple of the starting median province; Figure 9) shows the optimizer responding to where the "enough" line is drawn — but not in the way one might first guess. As θ rises from 0.50 to 1.10 the floor stays high (+14.7% → +16.5% over baseline) and inequality stays low (Gini 0.349 → 0.363, well below the baseline 0.439); what moves sharply is the *GDP cost*, from -12.0% to -1.5%. The mechanism is instructive: a low bar marks only the very poorest as "not enough," which the optimizer meets with concentrated, costly transfers; a higher bar implicates most provinces, which it meets with a broader and far cheaper lift. Where the sufficiency line sits changes less about the floor reached than about how much growth is sacrificed to hold it — the sufficientarian trade-off, made selectable rather than smuggled into a default. That the threshold is exposed as a slider rather than a hard-coded constant is the point: the one value that carries the entire moral content of the framework is placed in the user's hands, not smuggled into a default.

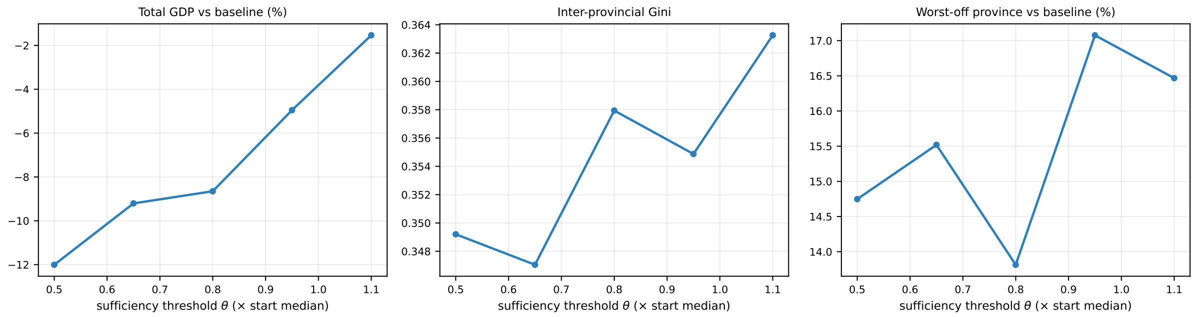

Figure 9: Sufficientarian sweep: final-year total GDP (left), Gini (center), and the worst-off province (right) as the sufficiency threshold θ rises. Means over 3 seeds. Raising the "enough" line shifts the optimizer's effort toward the floor.

9.9 How much the sign review matters: a propagation ablation

The inter-indicator propagation was changed twice in this work: from sign-*less* (every target moved with its source) to *documented* (polarity-derived signs), and then to *reviewed* (those signs corrected by the panel; Section 4.4). Table 6 re-runs the headline comparison (3 seeds) under all three to show the change is load-bearing, not cosmetic. The decisive step is making propagation sign-*aware* at all: moving from `signless` to `documented` shifts the egalitarian floor and Gini materially, because mis-signed links like income→poverty were previously pushing inequality the wrong way. The panel `review` on top — flipping 2 edges and dropping 3 unsupported ones — is a smaller, data-driven refinement. The control column (utilitarian Gini, which should barely move since utilitarianism is distribution-blind) stays nearly flat across modes, as expected.

Propagation mode	Egal. floor	Egal. Gini	Rawls floor	Util. Gini (control)
Sign-less (original)	+16.7%	0.417	+16.6%	0.436
Documented signs	+18.6%	0.408	+16.2%	0.435
Reviewed (ships)	+17.8%	0.406	+16.2%	0.435

Table 6: Propagation-mode ablation (3 seeds): floor change vs. baseline and final Gini under each inter-indicator propagation rule. Making propagation sign-aware (sign-less → documented) is the load-bearing change; the panel review is a smaller refinement on top. The utilitarian Gini is a control and barely moves.

9.10 The price of transparency, measured

Section 6 claims the explainability spectrum lets the black-box-vs-white-box performance gap be *measured* per problem rather than assumed — but the comparison so far uses only the neural policy. Training every model under one objective (utilitarian), one budget, and 3 seeds closes that gap (Table 7, Figure 10). The result is the encouraging one for contestability: in this twin the white-box models pay little or nothing for their legibility. The linear policy — the same per-province setup as the neural one minus the hidden layer — scores 96.2% of the neural policy's cumulative objective; the cross-entropy uniform and cluster policies and the rule list land within a few percent of it too. More strikingly, the gray-box Gaussian-process policy *edges past* the black box (101.7%), so here legibility is not merely cheap but, for one model, free. With the twin's effects bounded by the realism caps, the extra expressive capacity of a hidden layer buys little, and the inherently interpretable models recover essentially all of it — the empirical half of Rudin's argument [20], answered with data for this problem instance rather than asserted.

Policy model	Badge	Score (% of neural)	Total GDP	Gini
Neural (MLP)	black box	100.0	320.0	0.432
Linear	white box	96.2	298.4	0.456
Rule list	white box	95.6	295.0	0.445
Regional clusters	white box	93.6	283.3	0.433
Uniform (CEM)	white box	97.7	306.0	0.439
Uniform (Bayesian)	gray box	101.7	328.5	0.439

Table 7: Price of transparency (3 seeds, utilitarian objective): each policy model's cumulative objective score as a percentage of the neural policy's, with its explainability badge and final-year metrics. The white-box models recover most of the black box's performance, so transparency is cheap in this twin.

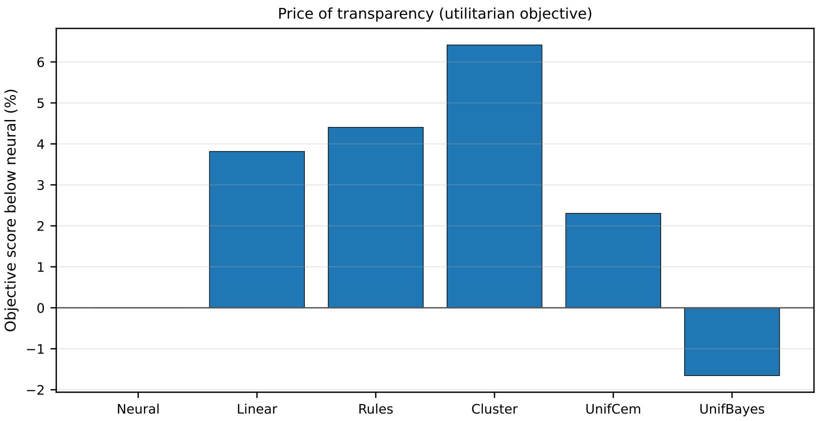

Figure 10: The score gap to the neural policy for each model (utilitarian objective, 3 seeds). Shorter bars are cheaper transparency; the white-box models give up little.

9.11 Reading the results

Six observations, with the standing caveat that they describe the simulator (Section 8.2) and are means over 6 seeds:

1. **The frameworks genuinely diverge.** With the model, data, budget, and search budget held fixed, swapping the per-year welfare function alone produces measurably different futures on all three evaluation axes (Table 4): total GDP ranges from -8.6% (maximin) to +11.1% (utilitarian) versus baseline, the Gini from 0.370 to 0.437, and the floor from +8.7% to +19.3%. With 6 seeds these carry tight bootstrap confidence intervals — the utilitarian GDP gain is +11.1±2.4%, its floor +8.7±3.9% (95% CI) — and the headline orderings below that we flag as robust are exactly those whose intervals do not overlap. The ethical choice is not decorative: it is the difference between the rows.

2. **The utilitarian optimum is a rising tide that under-serves the floor.** Utilitarianism grows total GDP the most (+11.1%) and leaves the Gini near baseline (0.437 vs. 0.439): the twin's multiplicative, capped dynamics make the sum-maximizer lift nearly every province

by a similar amount (Figure 4, a tight box around +11%). But distribution-blindness shows where it counts — the floor it reaches (++8.7%) is the *lowest* of the interventions and more seed-variable than the maximin-family floors (±11.3 on the worst-off province vs. ±7.0 for maximin and ±5.9 for CVaR). The rising tide does lift the floor, but only incidentally, and the frameworks that *target* it reach 8–10 points higher (below). The classic objection survives, in a milder form than the textbook: not sacrifice of the few, but indifference — the worst-off rise as a by-product rather than a goal.

3. **Maximin reliably raises the floor — and multi-seed replication is what shows it.** Under 6 seeds, maximin lifts the worst-off province by +17.8% (to 271.0) with one of the *lowest* floor-variances of any framework (±7.0, second only to its smoothed CVaR variant below), comfortably above the utilitarian floor — the difference principle doing what it promises. This is worth dwelling on methodologically: a *single*-seed version of this same experiment had shown maximin's floor landing fractionally *below* the utilitarian one, which invited a tidy "maximin under-optimizes its own objective" story; replication dissolves that story as seed luck (the utilitarian floor's huge variance is what produced the crossing). It is a small but clean demonstration of why the multi-seed protocol of Section 10.4 matters: the headline of a gradient-free comparison can invert between seeds. What *is* robust across seeds is the leveling-down *shape*: maximin still cuts the richest tail hard (down roughly 35–45% in Figure 4) because everything above the minimum is worth nothing to it — and the GDP it concedes for the higher floor is a real -8.6%.

4. **Smoothing maximin weakly dominates it — answering the sparsity question.** The CVaR objective, scoring the worst 20% of provinces instead of the single poorest, was included to test whether maximin's sparse one-province signal was the real source of its troubles. Across 6 seeds CVaR matches or beats hard maximin on every axis of the means: an equal-or-higher floor (272.8 vs 271.0), less GDP forgone (-4.2% vs -8.6%), lower run-to-run variance, and the *lowest Gini of any framework* (0.370). The honest reading, given the spreads: the *floor* difference between the two is not significant (the paired 95% bootstrap CI on the per-seed difference includes zero — they reach statistically the same floor), so smoothing the sparse signal does not *cost* the floor; what it robustly buys is a lower GDP price, tighter inequality, and less seed variance. That is still a clean answer to the sparsity question — broadening maximin's signal to the bottom quintile is a free lunch on every axis except the floor it leaves untouched — and a concrete argument for preferring the CVaR form in practice.

5. **Sen welfare is the efficiency–equity sweet spot.** The egalitarian run reaches the *highest floor of any framework* (+19.3%, though at the *largest* floor-variance of any framework, ±13.8) *and* still grows total GDP (+8.8%), holding the Gini at 0.406 — achieved (Figure 4) by lifting almost every province while trimming only a handful of the richest. Because its score multiplies growth by equality, it draws training signal from the whole distribution, which is plausibly why it alone pairs the top floor with positive growth. Its floor advantage over the utilitarian run is significant across 6 seeds (paired 95% bootstrap CI). The wellbeing run lands with the utilitarian on GDP (+9.9%) but distributes its gains tightly across every province, routed through its health/employment/poverty channels rather than GDP alone.

6. **The frontier exposes where the trade-off actually lives.** The uniform-lever Pareto search finds *no* efficiency–equity trade-off — efficiency and the floor co-move at correlation 1.000 and the Gini is flat to 10^{-6} (Figure 6). Giving the same search per-cluster levers (Figure 7) makes the Gini vary for the first time (0.411–0.418) — modest, and on an admittedly under-resolved front of 3 points, but *qualitatively absent* from the uniform run. Tellingly, GDP and the floor still co-move even on the clustered front: what targeting opens

is an efficiency-versus-*inequality* trade (hold back the richest cluster for a lower Gini), not an efficiency-versus-floor one. Equity, in this twin, is purchased by *targeting*, not by the national policy level. For decision support the implication is concrete: "how interventionist should national policy be?" is the wrong equity question; "who gets treated differently?" is the right one — and that is precisely what the contestability machinery (Section 7) exists to make auditable.

Two structural qualifications. First, the effect sizes are bounded by design: the policy-signal cap, the causal-rule strength, and the $\pm 6\%$/year realism cap mean that ten simulated years cannot diverge arbitrarily, so the inter-framework differences are modest in absolute terms — which is realistic for a decade of fiscal policy, and a direct consequence of the calibration discussed in Section 10. Second, the ES training budget (6 iterations) is the UI's interactive default, not a converged optimum, and 6 seeds tighten but do not eliminate run-to-run noise (the standard deviations in the text are non-trivial); each row should be read as "what this framework finds with the standard interactive budget, averaged over a handful of seeds," not as its global optimum.

10 Discussion

10.1 What the comparison shows

The experiment of Section 9 closes the loop the architecture promises: with everything held fixed except the per-year welfare function, the same optimizer is pulled to measurably different futures, and the frameworks' *relative signatures* are exactly as the theories say — maximin and its CVaR cousin reliably raise the floor while cutting the richest tail, Sen welfare reaches the highest floor of all while still growing, the wellbeing composite reroutes gains through non-GDP channels, and the distribution-blind utilitarian run grows the average the most while leaving the worst-off lowest and more seed-variable than the frameworks that target it. Two lessons generalize beyond the numbers. The first is methodological and was bought precisely by the multi-seed protocol: a *single*-seed version of this experiment had maximin's floor landing below the utilitarian one — a tidy, publishable "maximin under-optimizes its own objective" story that *evaporated under replication* as the utilitarian floor's large variance, a cautionary tale about reading a gradient-free optimizer's headline off one run. The second is substantive: a moral principle's practical meaning depends on the response structure of the system it optimizes *and* on how optimizable its objective is — which is why we added the smoothed CVaR maximin, and why the lesson it teaches (broadening maximin's one-province signal to the bottom quintile improves the growth and inequality it delivers at no cost to the floor) could only be read off an experiment that held the theory fixed. The methodological content is the point: *the moral criterion is an experimental variable*, and a system built this way can answer "what would the Rawlsian recommendation cost, and who pays?" with a number, a spread, and a map instead of a position paper.

10.2 Trade-offs exist only under calibrated scarcity

An engineering finding that is an ethical finding in disguise: with looser caps (e.g., a policy-signal cap of 0.5 instead of 0.03) every intervention saturates the $\pm 6\%$ growth limit and all growth-favoring objectives collapse onto the same optimum — the ethical comparison becomes vacuously flat. The frameworks differentiate only because the budget constraint and the effect caps force genuine choices. Distributive justice presupposes scarcity, in the model exactly as in philosophy; a simulator that lets every policy succeed everywhere has abolished the subject matter.

10.3 The assumptions dominate aggressive scenarios

Since the ML layer's relative response saturates its cap, the differentiated structure of *which* levers move *which* indicators comes mostly from the hand-written causal rules. The sensitivity band exists precisely because of this: it shows, per run, how much of the projection rests on those hand-tuned assumptions. The honest reading of Section 9 is therefore: the *direction* and *ordering* of the inter-framework differences are meaningful (they follow from the objectives' mathematics interacting with documented causal structure); the *magnitudes* are hostage to constants a domain expert wrote down.

10.4 Threats to validity

Beyond the simulator-versus-reality category error (stated throughout), three internal threats deserve note. (i) *Search noise and signal sparsity*: gradient-free training with small budgets has run-to-run variance. The results above already mitigate this by replicating every framework over 6 seeds and reporting mean $\pm$ standard deviation (the bands in Figure 2); the framework *ordering* is stable across seeds where the bands do not overlap, but a few-seed sample still cannot fully pin the magnitudes, and more seeds would tighten them further. The objectives also differ in how much training signal they emit per rollout — maximin scores one province in 110, Sen welfare the whole distribution — which is exactly why the smoothed CVaR objective is included: it isolates that asymmetry by holding the Rawlsian *theory* fixed while scoring the worst 20% of provinces, so the gap between the `rawlsian` and `cvar` rows of Table 4 is a direct, in-experiment estimate of how much of hard maximin's poor floor is optimization difficulty rather than the difference principle itself. (ii) *Objective scale effects*: the objectives have different score scales, and although each optimizer only compares candidates within one objective, ES step-size interacts with score curvature, so frameworks may be optimized with effectively different aggressiveness (the wellbeing run's large relative training improvement, +72.2%, is mostly an artifact of its near-zero starting score). (iii) *Proxy leakage*: all three evaluation axes are GDP-denominated, which understates what the wellbeing framework achieves on its own currency (health, employment, poverty) — its row in Table 4 is scored in a currency it deliberately deprioritizes.

10.5 Limits of the paradigm

Finally, the optimization paradigm itself has ethical limits that no objective choice cures. Relational equality (Section 5) resists encoding as a function over indicator vectors. Procedural justice — legitimacy, participation, consent — is invisible to end-state scoring. People missing from official statistics are missing from every objective (Section 4.2). And the unit-of-fairness choice (the province) hides all within-province distribution. SERA's contribution is not to solve these but to keep them written on the box.

11 Conclusions

SERA demonstrates, end to end, what it takes to put an explicit ethical governance layer around an AI optimizer that advises on public-resource allocation. The central design move is separation of concerns: the world model, the search procedure, and the moral criterion are three independent, swappable components, so the value judgment is forced out of the code and into a visible, user-facing choice. Around that core, the project assembles the scaffolding responsible deployment would demand — an explainability spectrum with a measured price of transparency, post-hoc audits that state their own fidelity, sensitivity bands that quantify how much rests on assumptions, a human-in-the-loop candidate/adopt workflow, and the model-card/datasheet/ethics documentation triplet mapped onto the EU AI Act's high-risk obligations.

The experimental comparison supports the project's framing claim — there is no single "correct" ethical algorithm for resource allocation — and sharpens it in two ways the design did not anticipate. First, a moral principle's practical meaning depends both on the dynamics it optimizes and on how readily it can be optimized: maximin and its smoothed CVaR variant reliably protected the worst-off province (at a modest GDP cost and by visibly sacrificing the richest tail), while the distribution-blind utilitarian run grew the average most but left the floor lowest and more seed-variable than the maximin-family frameworks that target it. That last comparison only became trustworthy under replication: a single seed had shown maximin's floor falling *below* the utilitarian one — a tidy "maximin under-optimizes itself" story that multi-seed replication exposed as search luck. The methodological moral is as durable as the philosophical one: the headline of a gradient-free ethical comparison can invert between seeds, so replication is not optional housekeeping but part of the claim. Second, the locus of the ethical choice is not where the dropdown suggests: uniform national policy moved efficiency and the floor in lockstep with inequality untouched, so the entire distributional question lives in per-province targeting — exactly the part of a policy that contestability machinery must make auditable. What an ethically serious system can do is not resolve the choice of framework but *refuse to make it silently* — presenting trade-offs to be inspected, with every candidate's reasoning auditable and every assumption priced.

The honest limitations are equally part of the conclusion: the twin's causal core is hand-written and dominates aggressive scenarios; the experimental magnitudes are bounded by calibrated caps; the data measure only the measured population; and the unit of fairness is the province. A system that knows and says what it cannot see is the difference between decision support and automation bias with extra steps. Future work follows directly: a user-thresholded sufficientarian objective and a concavity-parameterized prioritarian one (Section 5.9), multi-seed replication of the framework comparison, population-weighted welfare variants, and within-province distribution if data permits.

12 Reproducibility

The project requires Python 3.11 and Node.js. From the repository root:

```
python -m venv .venv
.venv\Scripts\activate          # Windows (macOS/Linux: source .venv/bin/activate)
pip install -r requirements.txt
```

Download the data (rate-limited; the full set is slow), train the twin, and run the tests:

```
$env:PYTHONPATH = "$pwd\src"
python src/sera/downloader.py --indicator all
python -m sera.twin.cli --mode train-and-simulate --model-type ridge --sim-years 5
python -m pytest tests -q
```

This writes `twin_models.joblib` (used by the UI) and `simulation_results.csv`. The experiments of Section 9 are reproduced by (matplotlib required for the figures):

```
.venv\Scripts\python.exe report\run_experiments.py        # headline comparison + Pareto
.venv\Scripts\python.exe report\run_prioritarian_sweep.py # the continuum sweep
.venv\Scripts\python.exe report\run_ablation.py           # propagation-mode ablation
.venv\Scripts\python.exe report\run_transparency.py       # price of transparency
.venv\Scripts\python.exe report\run_sufficientarian_sweep.py  # sufficiency-threshold sweep
.venv\Scripts\python.exe report\plot_results.py           # figures + results_macros.tex
```

which write the raw JSON to `report/results/`, the figures to `report/figures/`, and every number quoted in this report to `report/results_macros.tex`. Then launch the control room:

```
cd ui
npm install
npm start
```

The UI exposes the policy-model and ethical-objective dropdowns in the header, the per-province allocators and budget meter, the optimizer with its three candidates and sensitivity band, the `compare-objectives` equity dashboard, and the `pareto-front` efficiency–equity frontier. The app makes no network requests at runtime; raw ISTAT data is not redistributed and is fetched by each user under ISTAT's terms.

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A Implementation Algorithms

Since the algorithms are central to how the ethical objectives act on the system, the main ones are listed here for reference.

Algorithm 1: One simulated year (`DigitalTwinSimulator.simulate_year`)

Data: state s_t (110 provinces × indicators), levers a_t, trained models $\{f_k\}$, caps κ_{sig}=0.03, κ_{rule}=0.02, $g_{\max}$=0.06
Result: next-year state s_{t+1}

foreach *indicator k, province p* **do**
  $\hat{y} \leftarrow f_k(\text{lagged indicators}, a_t)$; $\hat{y}_0 \leftarrow f_k(\text{lagged indicators}, a_{\text{baseline}})$;
  $\delta \leftarrow \text{clip}\big((\hat{y} - \hat{y}_0)/|y_t|, -\kappa_{\text{sig}}, \kappa_{\text{sig}}\big)$; `// anchored: only the policy signal is trusted`
  $y' \leftarrow y_t \cdot (1 + \delta)$;
  foreach *lever ℓ with a causal edge $\ell \to k$* **do**
    $\sigma \leftarrow \tanh\big((a_{t,\ell} - \text{median}_\ell)/\text{scale}_\ell\big)$; `// historical normalization`
    $y' \leftarrow y' \cdot \big(1 + \kappa_{\text{rule}} \cdot \sigma \cdot \text{dir}(\ell, k)\big)$;
propagate inter-indicator effects along the dependency graph (dampened);
clip each indicator to its logical bounds;
cap year-over-year change at $\pm g_{\max}$ per indicator ; `// realism speed limit`
return s_{t+1};

Algorithm 2: National budget constraint (applied every simulated year)

Data: allocations a, state s, reserve r, spending intensity β

Result: feasible allocations a', new reserve r'

pool $\leftarrow \beta \cdot \mathrm{GDP}(s) \cdot \overline{\left(a_\tau / a_\tau^{\mathrm{base}}\right)}_{\tau \in \text{tax levers}}$; `// tax cuts below baseline shrink the pool`

cost $\leftarrow \sum_p \sum_{\ell \in \text{spending}} \mathrm{cost}\big(a_{p,\ell}\big)$;

if $\mathit{cost} > \mathit{pool} + r$ **then**

  scale all spending levers down by $(\mathrm{pool} + r)/\mathrm{cost}$ toward baseline;

else

  leave allocations unchanged;

$r' \leftarrow \max\big(0,\ r + \mathrm{pool} - \min(\mathrm{cost},\ \mathrm{pool} + r)\big)$; `// unspent budget carries over`

return (a', r');

Algorithm 3: Mirrored-sampling Evolution Strategies (`neural` and `linear` policies)

Data: evaluate $J(\theta)$ (cumulative ethical welfare, Eq. 2), init θ_0, popsize N=6, noise σ=0.25, step α=0.2

$(\theta^*, J^*) \leftarrow (\theta_0, J(\theta_0))$;

for $i = 1 \ldots \mathit{iterations}$ **do**

  sample $\varepsilon_1, \ldots, \varepsilon_N \sim \mathcal{N}(0, I)$; evaluate J at $\theta \pm \sigma\varepsilon_j$ ($2N$ rollouts, mirrored);

  normalize scores to advantages A (zero mean, unit variance);

  $g \leftarrow \frac{1}{2N\sigma} \sum_j A_j\, \varepsilon_j^{(\pm)}$; `// ES gradient estimate`

  $\theta \leftarrow \theta + \alpha g$; update (θ^*, J^*) if any candidate improved;

return θ^* ; `// best evaluated weights, not the last center`

Algorithm 4: Cross-Entropy Method (shared by `rules`, `cluster_cem`, `uniform_cem`)

Data: evaluate $J(\cdot)$ (cumulative ethical welfare, Eq. 2), start $x_0 \in [0,1]^d$, popsize N, elite fraction ρ, smoothing α, σ_{init}=0.25, σ_{floor}=0.02

$\mu \leftarrow x_0$; $\sigma \leftarrow \sigma_{\mathrm{init}}\mathbf{1}$; $(x^*, J^*) \leftarrow (x_0, J(x_0))$;

for $i = 1 \ldots \mathit{iterations}$ **do**

  sample N candidates $x_j \sim \mathrm{clip}\big(\mathcal{N}(\mu, \mathrm{diag}(\sigma^2)),\ 0, 1\big)$ and score $J(x_j)$;

  $E \leftarrow$ top $\lceil \rho N \rceil$ candidates by score;

  $\mu \leftarrow \alpha\, \mathrm{mean}(E) + (1-\alpha)\, \mu$; $\sigma \leftarrow \max\big(\alpha\, \mathrm{std}(E) + (1-\alpha)\, \sigma,\ \sigma_{\mathrm{floor}}\big)$;

  update (x^*, J^*) if improved;

return x^*;

Algorithm 5: Efficiency–equity Pareto front (`sera.twin.pareto`)

Data: rollout env, population size N, generations G

initialize population P of uniform lever vectors in $[0, 1]^{20}$;
for $g = 1 \ldots G$ **do**
  foreach $x \in$ *offspring of P (SBX crossover + polynomial mutation)* **do**
    roll out x through the twin; record $\left(\sum_p y_p,\ -G(y),\ \min_p y_p\right)$ of the final simulated year;
  fast non-dominated sort of $P \cup$ offspring; select next P by front rank, then crowding distance;
tag the extreme points of the first front as the utilitarian ($\max \sum y$), egalitarian ($\min G$), and Rawlsian ($\max \min y$) corners;
return the non-dominated front with every point's lever table;

Algorithm 6: Post-hoc audit of the neural policy (`sera.twin.explain`)

Data: trained policy π, starting state s_0, feature set F
Result: permutation importances; surrogate tree with fidelity

$D \leftarrow$ decision matrix of π on s_0 ; `// (provinces × levers), lever-range units`
foreach *indicator* $k \in F$ **do**
  for $r = 1 \ldots n_{repeats}$ **do**
    shuffle column k of s_0 across provinces; recompute decisions D';
    record $\text{shift}_r \leftarrow \text{mean}\,|D' - D|$;
  $\text{importance}(k) \leftarrow \text{mean}_r(\text{shift}_r)$;
split provinces into train/held-out; fit a shallow decision tree T : indicators $\rightarrow$ levers on π's decisions;
fidelity $\leftarrow R^2$ of T on the held-out provinces' decisions ; `// reported next to the tree, not assumed`
return importances (sorted), T, fidelity;